

46 YEARS OF EXCELLENCE | NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR.IIT_*CO-SC(MODEL-A& APEX) **GTM-12** **Date: 06.01.2025**
Time: 3.00Hrs. **JEE MAIN MODEL** **Max. Marks: 300**

PHYSICS MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

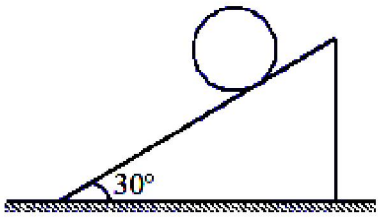
1. The electric field inside a sphere having charge density related to the distance from the centre as $\rho = \alpha r$ (α is a constant) is

A) $\frac{\alpha r^3}{4\epsilon_0}$ B) $\frac{\alpha r^2}{4\epsilon_0}$ C) $\frac{\alpha r^2}{3\epsilon_0}$ D) none of these

2. A uniform disc of mass 2 kg and radius 50 mm is released from rest on a smooth fixed incline plane as shown. If the length of incline equals 160 cm, the time taken by the disc to arrive at bottom equals (Take: $g = 10\ ms^{-2}$)

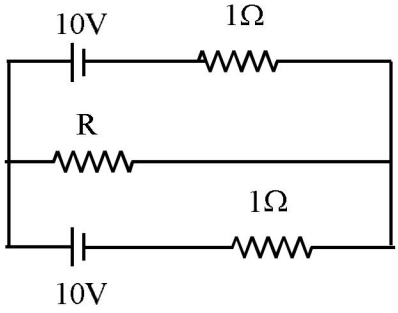
APEX JEE-MAIN 2025

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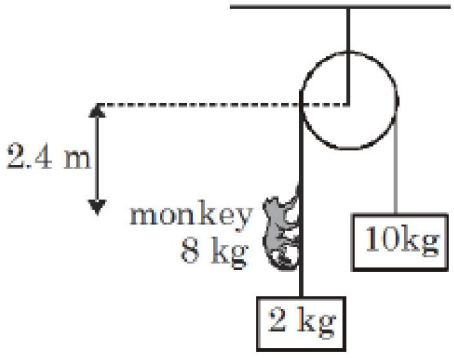
A) 1 s B) 0.8 s C) 0.6 s D) 0.5 s

3. Maximum power developed across variable resistance R in the circuit shown in figure is



A) 50 W B) 75 W C) 25 W D) 100 W

4. Two blocks of mass 10 kg and 2 kg respectively are connected by an ideal string passing over a fixed smooth pulley as shown in figure. A monkey of 8 kg started climbing the string with constant acceleration 2 ms^{-2} with respect to string at $t = 0$. Initially the monkey is 2.4 m from the pulley. Find the time taken by the monkey to reach the pulley.



A) 1 sec B) 2 sec C) 4 sec D) 8 sec

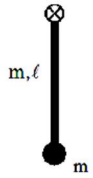
5. Which of the following transitions of He^+ ion will give rise to spectral line which has same wavelength as some spectral line in hydrogen atom?

A) $n = 4$ to $n = 2$ B) $n = 6$ to $n = 5$ C) $n = 6$ to $n = 3$ D) None of these

APEX JEE-MAIN 2025

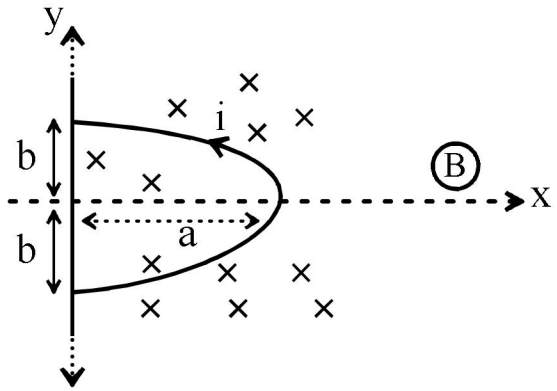
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6. A particle is attached to the lower end of a uniform rod which is hinged at its other end as shown in the figure. Another identical particle moving horizontally, collides inelastically and sticks to it. The minimum speed of moving particle so that the rod with particle performs circular motion in a vertical plane will be [length of the rod is ℓ, consider masses of both particles and rod to be same]



A) $\sqrt{10g\ell}$ B) $\sqrt{20g\ell}$ C) $\sqrt{\frac{70}{3}g\ell}$ D) $\sqrt{\frac{175}{3}g\ell}$

7. In the figure, there is a conducting wire having current i and which has a shape of closed half ellipse $\left[\frac{x^2}{a^2}+\frac{y^2}{b^2}=1\right]$ is kept in a uniform magnetic field B as shown. The magnitude of magnetic dipole moment of loop and torque acting on it are



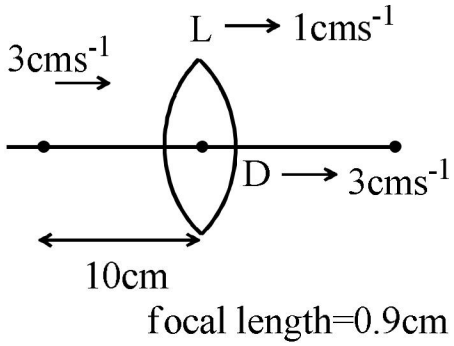
A) $\pi\ abi,\ 0$ B) $\pi\ ab\ i,\ \pi\ abiB$

C) $\frac{\pi\ ab\ i}{2},\ \frac{\pi\ abiB}{2}$ D) $\frac{\pi\ ab\ i}{2},\ 0$

8. The figure shows the initial position of a point source of light s, a detector D and lens L. Now at t =0, all three starts moving towards right with different velocity as shown in figure. The times at which detector receives the maximum light, is (Assume that the detector is initially just touching the lens).

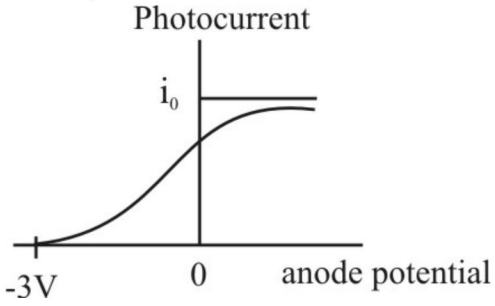
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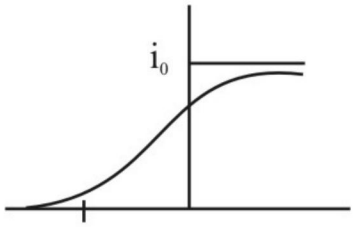
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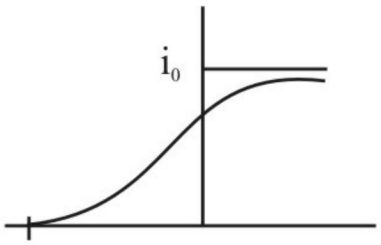


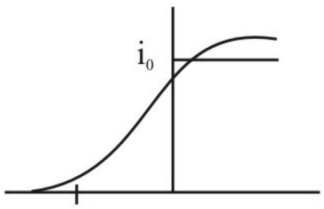
A) 5 s and 10 s B) 0.5 s and 4.5 s C) 2 s and 3 s D) Only 10 s

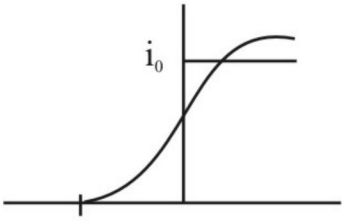
9. Figure shows the photo-current *vs* anode potential graph when photons of 6 eV are incident on cathode in a photoelectric experiment set-up. In the same experimental set-up, if photons of 8 eV and same intensity are incident on cathode, then anode-potential *vs* photo-current graph will be (assume 100% efficiency of photons to eject photo-electrons in both cases)



A) 

B) 

C) 

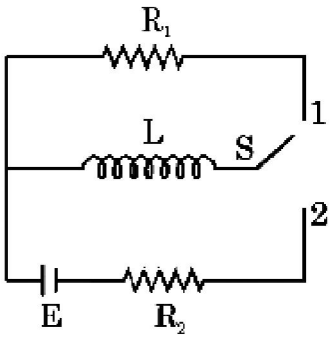
D) 

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10. A sound source is located somewhere along the x-axis. Experiment shows that the same wave front simultaneously reaches listeners at $x = -10$m and $x = +4.0$ m. A third listener is positioned along the positive y-axis. What is y-coordinate (in m) if the same wave front reaches at him at the same instant it does the first two listeners?

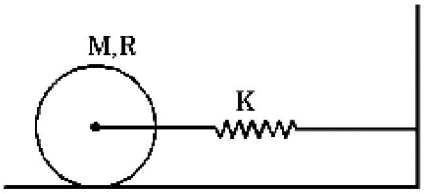
A) $\sqrt{40}$ B) 3 C) 7 D) $\sqrt{30}$

11. In the circuit shown switch S is connected to position 2 for a long time and then joined to position 1. The total heat produced in resistance R_1 is



A) $\frac{LE^2}{2R_2^2}$ B) $\frac{LE^2}{2R_1^2}$ C) $\frac{LE^2}{2R_1R_2}$ D) $\frac{LE^2(R_1+R_2)^2}{2R_1^2R_2^2}$

12. A solid cylinder attached to horizontal massless spring can roll without slipping along horizontal surface. Find time period of small oscillation.



A) $2\pi\sqrt{\frac{M}{2k}}$ B) $\pi\sqrt{\frac{3M}{2k}}$ C) $\pi\sqrt{\frac{2M}{3k}}$ D) $2\pi\sqrt{\frac{3M}{2k}}$

13. The minimum and maximum distance of a satellite from the centre of the earth are $2R$ and $4R$, respectively, where R is the radius of earth and M is the mass of earth. The radius of curvature at the point of maximum distance is

A) $\frac{8R}{3}$ B) $\frac{4R}{3}$ C) $\frac{3R}{8}$ D) $\frac{3R}{4}$

APEX JEE-MAIN 2025

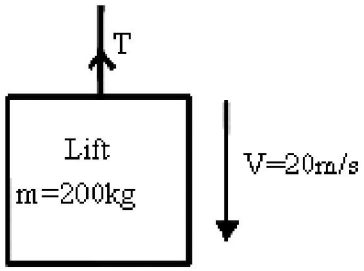
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14. Force $F = \frac{x}{\rho} + y$, where ρ is mass density then dimensional formula of x and y respectively are

A) $[M^2\ L\ T^{-2}]$, $[M^0\ L^1\ T^0]$ B) $[M^1\ L^1\ T^{-2}]$, $[M^2\ L^2\ T^{-2}]$

C) $[M^2\ L^{-2}\ T^{-2}]$, $[M^1\ L^1\ T^{-2}]$ D) $[M^1\ L^2\ T^{-2}]$, $[M^1\ L^1\ T^{-2}]$

15. If this lift is moving downward with constant velocity 20 m/s, find work done by tension in 2 sec

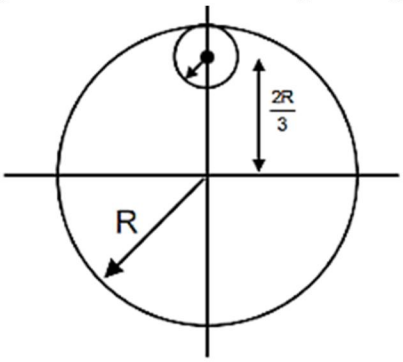


A) -8×10^4 J B) -20×10^4 J C) -4×10^4 J D) -12×10^4 J

16. A particle is projected at $t = 0$ with velocity u at angle θ with the horizontal. Then the ratio of the tangential acceleration and the radius of curvature at the point of projection is

A) $\frac{g^2 \sin 2\theta}{u^2}$ B) $\frac{2g^2 \sin\theta}{u^2}$ C) $\frac{g^2 \sin\theta}{u^2}$ D) $\frac{g^2 \sin 2\theta}{2u^2}$

17. From a uniform circular disc of radius R and mass9M, a small disc of radius R/3 is removed as shown in the figure. The moment of inertia of the remaining disc about an axis perpendicular to the plane of the disc and passing through centre of disc is



A) $\frac{40}{9}MR^2$ B) $10\ MR^2$ C) $\frac{37}{9}MR^2$ D) $4\ MR^2$

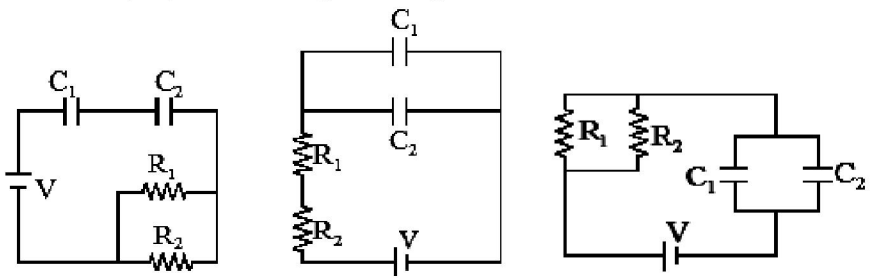
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18. A spherical solid ball of volume V is made of a material of density ρ_1. It is falling through a liquid of density $\rho_2 (\rho_2 < \rho_1)$. Assume that the liquid applies a viscous force on the ball that is proportional to the square of its speed v, *i.e.*, $F_{\text{viscous}} = -kv^2 (k > 0)$. Then terminal speed of the ball is

A) $\sqrt{\frac{V g(\rho_1 - \rho_2)}{k}}$ B) $\frac{V g \rho_1}{k}$ C) $\sqrt{\frac{V g \rho_1}{k}}$ D) $\frac{V g(\rho_1 - \rho_2)}{k}$

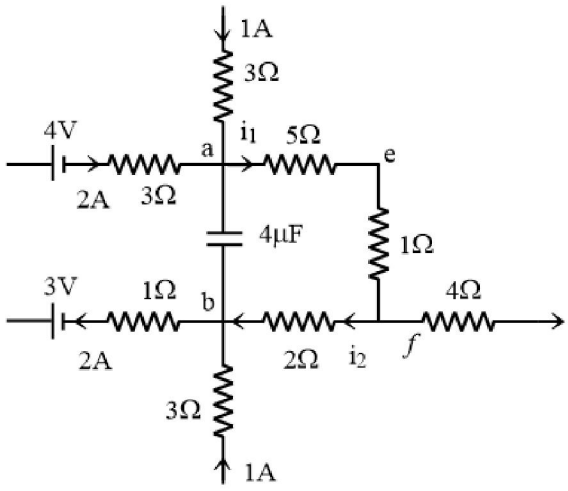
19. Given $R_1 = 1\Omega$, $R_2 = 2\Omega$, $C_1 = 2\mu F$, $C_2 = 4\mu F$. The time constants (in μS) for the circuits I, II, III are respectively



(I) (II) (III)

A) 18, $\frac{8}{9}$, 4 B) 18, 4, $\frac{8}{9}$ C) 4, $\frac{8}{9}$, 18 D) $\frac{8}{9}$, 18, 4

20. A part of circuit in steady state along with the currents flowing in the branches, the value of resistances etc., is shown in figure. Calculate the energy stored in the capacitor



APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

A) 8×10^{-1} J B) 8×10^{-2} J C) 8×10^{-3} J D) 8×10^{-4} J

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

21. A massless metal sheet is placed on a horizontal table top lubricated with oil. The sheet is a square of side length $\ell = 1.0$m and the oil layer has thickness $h = 1.0$ mm. Initially one edge of the sheet coincides with one edge of the table. The sheet is pulled outwards without rotation with a constant force $F = 15$ N. If coefficient of viscosity of the oil is $\eta = 0.2$ N-s/m^2, how long(in second) will it take to pull half of the sheet out of the table?

22. A monochromatic light of wavelength 6429 Å is used in YDSE. Now both slits are covered by two thin slab of refractive index 3.2 and 2.7respectively. By doing so central bright fringe shifts to original 7thbright fringe. If both slabs have same thickness then thickness of slab is (in μm). (nearest integer)

23. In changing the state of a gas adiabatically from an equilibrium state A to another equilibrium state B, an amount of work equal to 22.17 J is done on the system. If the gas is taken from state A to B via a process in which the net heat absorbed by the system is 9.35 cal, how much is the net work done (in joule) by the system in the latter case?(Take 1 cal = 4.19 J) (nearest integer)

24. A transverse periodic wave on a string with a linear mas density of 0.200 kg/m is described by the following equation.

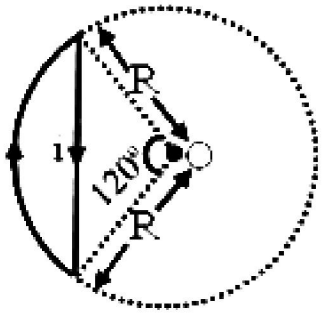
$$y = 0.05\sin(420t - 21.0x)$$

Where x and y are in metres and t is in seconds. Tension in the string is _____

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25. Net magnetic field at the centre O of circle due to current carrying loop as shown in figure is $\frac{\mu_0 i\left(x\sqrt{x}-\pi\right)}{2x\pi R}$ where x is



CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. The value of $\Delta_f G^0$ for formation of $Cr_2O_{3(s)}$ is $-540\,kJmol^{-1}$ and that of $Al_2O_{3(s)}$ is $-827\,kJmol^{-1}$. What is the value of $\Delta_r G^0$ for the reaction?

$$\frac{4}{3}Al_{(s)}+\frac{2}{3}Cr_2O_{3(s)} \rightarrow \frac{2}{3}Al_2O_{3(s)}+\frac{4}{3}Cr_{(s)}.$$

A) $-574\,kJ\,mol^{-1}$ B) $-191.33\,kJ\,mol^{-1}$ C) $+574\,kJ\,mol^{-1}$ D) $-287\,kJ\,mol^{-1}$

27. Which of the following statements is incorrect?

A) $[Ag(NH_3)_2]^+$ is linear with sp hybridized Ag^+.

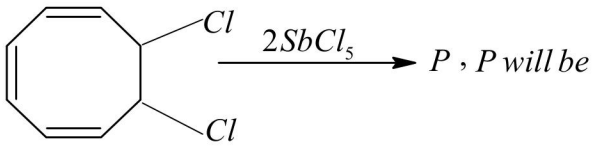
B) $NiCl_4^{2-}$, VO_4^{3-} and MnO_4^- have tetrahedral geometry.

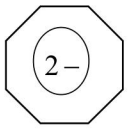
C) $[Cu(NH_3)_4]^{2+}$, $[Pt(NH_3)_4]^{2+}$ and $[Ni(CN)_4]^{2-}$ have square planar geometry.

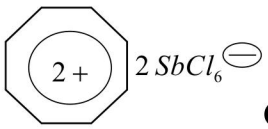
D) $Fe(CO)_5$ has trigonal bipyramidal structure with sp^3d hybridized iron.

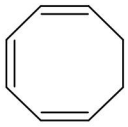
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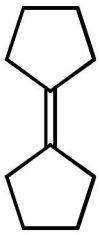
28. 

A) 

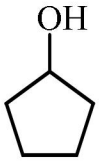
B) 

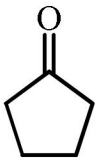
C) 

D) mixture of A and B

29. 

$\xrightarrow[Zn/H_2O]{O_3} A \xrightarrow[(or)\,LiAlH_4]{H_2/Ni} B \xrightarrow[\Delta]{H^+} C;$ here final product "C" is

A) 

B) 

C) 

D) 

30. The solubility of a specific non-volatile salt is 4g in 100g of water at 25^0C. If 2.0g, 4.0g and 6.0gm of the salt is added to 100g of water at 25^0C, in the systems X, Y and Z. The vapour pressure would be

A) $X < Y < Z$ B) $X > Y > Z$ C) $Z > X = Y$ D) $X > Y = Z$

31. 0.1M formic acid solution is titrated against 0.1M $NaOH$ solution, what would be the difference in pH between $\frac{1}{5}$ and $\frac{4}{5}$ stages of neutralization of acid?

A) $2\log\frac{3}{4}$ B) $2\log\frac{1}{5}$ C) $\log\frac{1}{3}$ D) $2\log 4$

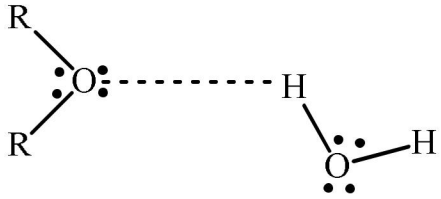
32. Given below are two statements one is labeled as Assertion (A) and other is labeled as Reason (R)

Assertion: The miscibility of ethers with water resembles those of alcohols of the same molecular mass, both ethoxyethane and butan-1-ol are miscible to almost the same extent i.e.,7.5 and 9 g per 100 mL water.

APEX JEE-MAIN 2025

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Reason : Just like butan-1-ol, oxygen of ethoxyethane can also form hydrogen bonds with water molecule as shown.



In the light of the above statements, choose the most appropriate answer from the options given below:

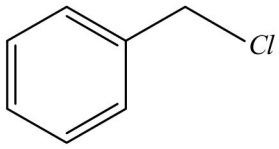
A) Both (A) and (R) are true and (R) is the correct explanation of (A).

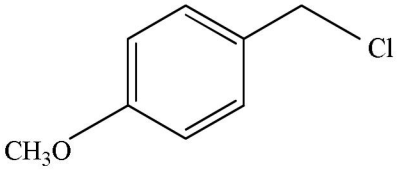
B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

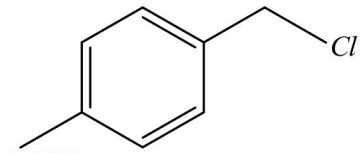
C) (A) is true but (R) is false.

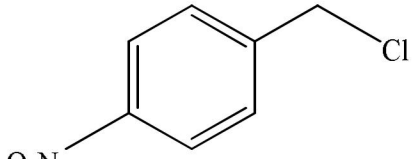
D) (A) is false but (R) is true.

33. Which of the following will react faster with Ag^+ ?

A) 

B) 

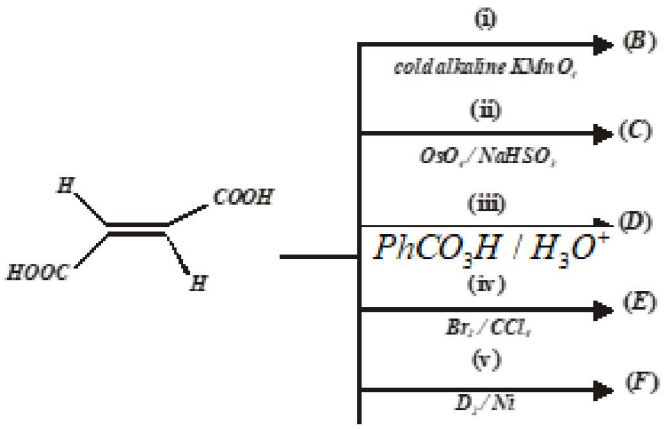
C) 

D) 

34. Which of the following reactions will yield racemic mixture?

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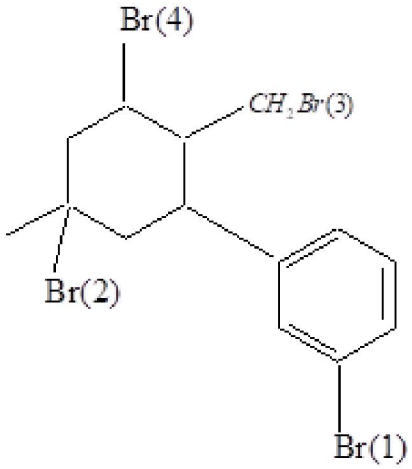
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A) (I), (III), (V) B) (I), (II), (V)

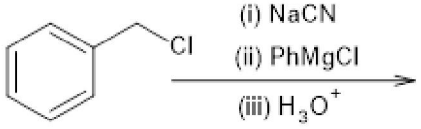
C) (I), (II), (IV) D) (III), (IV), (V)

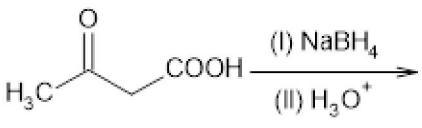
35. Arrange the below mentioned bromine atoms in decreasing order of their reactivity towards hydrolysis through S_N1 path



A) 2 >1 > 4 > 3 B) 2 > 1 > 3 > 4 C) 2 > 4 > 3 > 1 D) 4 > 2 > 1 > 3

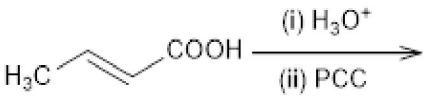
36. Identify the reactions where the final product contains ketone functional group.

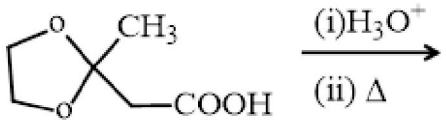
a) 

b) 

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c) 

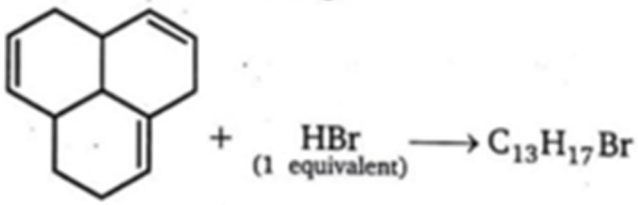
d) 

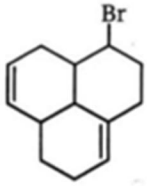
A) (a) and (d) only
B) (a), (c) and (d) only
C) (b), (c) and (d) only
D) (a), (b), (c) and (d)

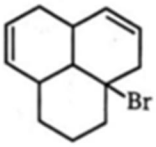
37. A 75 gm 4%$\left(\frac{W}{W}\right)$ aqueous NaOH solution is distilled for some time. The water collected has a mass of 25 gms. To this distilled solution X gms of KOH(s) is added, so that molality of OH^- ion in the resulting solution becomes 5. Find X?

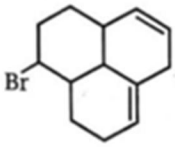
A) 6.62 g
B) 8.96 g
C) 17.92 g
D) 13.24 g

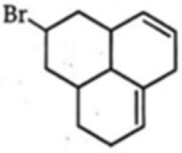
38. Which of the following bromides is the major product of the reaction shown below, assuming that there are no carbocation rearrangement?



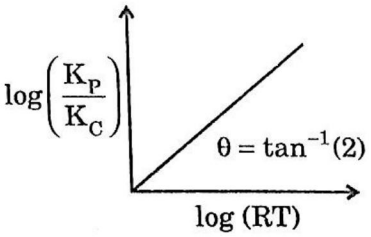
A) 

B) 

C) 

D) 

39. Which of the following reaction satisfies the given graph?



A) $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$
B) $NH_4HS_{(s)} \rightleftharpoons NH_{3(g)} + H_2S_{(g)}$

APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

C) $CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$ D) $CuSO_4.5H_2O_{(s)} \rightleftharpoons CuSO_4.2H_2O_{(s)} + 3H_2O_{(g)}$

40. An aqueous solution containing a non volatile solute is heated in an open vessel till the boiling point is reached. Thereafter, the boiling is continued and the temperature & the vapour pressure of the solution are recorded at 5 min, 10 min, and 15 min. The corresponding temperatures and pressures are T_5, T_{10}, T_{15} and P_5, P_{10}, P_{15}, respectively. The correct relationship among them is:

A) $T_5 < T_{10} < T_{15}$ and $P_5 = P_{10} = P_{15}$ B) $T_5 = T_{10} = T_{15}$ and $P_5 = P_{10} = P_{15}$

C) $T_5 = T_{10} = T_{15}$ and $P_5 < P_{10} < P_{15}$ D) $T_5 < T_{10} < T_{15}$ and $P_5 < P_{10} < P_{15}$

41. $P_4 + 3O_2 \rightarrow P_4O_6, \Delta H = -1500kJ$. Bond enthalpies of $O=O$, $P-O$ are $500kJmol^{-1}$ and $350kJmol^{-1}$ respectively. What is '$P-P$' bond energy?

A) $300kJmol^{-1}$ B) $200kJmol^{-1}$ C) $100kJmol^{-1}$ D) $400kJmol^{-1}$

42. The order of boiling points of hydrides of VIA group elements.

A) $H_2O > H_2Te > H_2Se > H_2S$ B) $H_2Te > H_2O > H_2Se > H_2S$

C) $H_2O > H_2S > H_2Se > H_2Te$ D) $H_2O < H_2S < H_2Se < H_2Te$

43. Match List-I with List-II

List-I		List-II	
(P)	$\psi_{MO} = \psi_A - \psi_B$	(I)	Dipole moment
(Q)	$\mu = Q \times r$	(II)	Bonding molecular orbital
(R)	$\frac{N_b - N_a}{2}$	(III)	Anti-bonding molecular orbital
(S)	$\psi_{MO} = \psi_A + \psi_B$	(IV)	Bond order

A) (P)-(II), (Q)-(I), (R)-(IV), (S)-(III)

B) (P)-(III), (Q)-(IV), (R)-(I), (S)-(II)

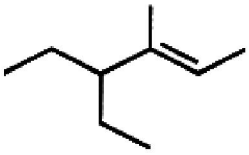
C) (P)-(III), (Q)-(I), (R)-(IV), (S)-(II)

APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

D) (P)-(III), (Q)-(IV), (R)-(II), (S)-(I)

44. The IUPAC name of the following compound is



A) 3-Ethyl-4-methylhex-4-ene
B) 4, 4-Diethyl-3-methylbut-2-ene
C) 4-Methyl-3-ethylhex-4-ene
D) 4-Ethyl-3-methylhex-2-ene

45. The difference between amylose and amylopectin is

A) Amylopectin have 1 → 4 α- glycosidic linkage and 1→ 6 α- glycosidic linkage

B) Amylose have 1 → 4 α- glycosidic linkage and 1→ 6 β- glycosidic linkage

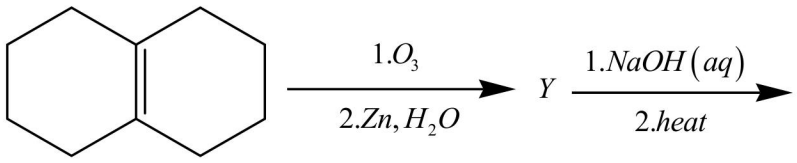
C) Amylopectin have 1 → 4 α- glycosidic linkage and 1→ 6 β- glycosidic linkage

D) Amylose is made up of glucose and galactose

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

46. In the scheme given below, the total number of sp^2 hybridised carbon atoms in the final major product is



47. A hydrogen gas electrode is made by dipping platinum wire in a solution of HCl of pH = 10 and by passing hydrogen gas around the platinum wire at one atm pressure. The oxidation potential of electrode would be in terms of $x \times 10^{-2}$ V. What is x?

(Give: $\frac{2.303RT}{F} = 0.059$)

APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

48. Total number of geometrical isomers for the complex $\left[RhCl(CO)(PPh_3)(NH_3)\right]$ is

49. An organic compound containing nitrogen with molecular mass 180 is acylated with CH_3COCl in the presence of pyridine to get a compound with molecular mass 390. The number of amino groups present per molecule of the former compound is____.

50. For a reaction $A \longrightarrow 2B + C$ the half-lives are 100 s and 50 s when the concentration of reactant A is 0.5 mol L^{-1} and 1.0 mol L^{-1} respectively. The order of the reaction is _____

MATHS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. The series of natural numbers is divided into groups as (1), (2, 3, 4), (3, 4, 5, 6, 7), (4, 5, 6, 7, 8, 9, 10),..... . The sum of elements in the 20th group is

A) 1368 B) $(38)^2$ C) $(39)^2$ D) 38×39

52. The lines $L_1 : x = y = z$, $L_2 : x = \frac{y}{2} = \frac{z}{3}$ and a line L_3 is passing through (1, 1, 1) form a triangle of area $\sqrt{6}$ units, (1, 1, 1) being one of the vertices of the triangle. Then the point of intersection of the L_3 with L_2 is

A) (1, 2, 3) B)(2, 4, 6) C)$\left(\frac{4}{3}, \frac{8}{3}, 4\right)$ D)(1, 5, 7)

53. The combined equation of 2 altitudes of an equilateral triangle is $x^2 - 3y^2 - 4x + 6\sqrt{3}y - 5 = 0$. The third altitude has equation

A) $x + 2 = 0$ B) $y = \sqrt{3}$ C) $x = 2$ D) None of these

APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

54. The mean marks of students of a school is 56. The mean marks of girls is 60 and that of boys is 50. If number of boys and girls are n and m respectively, then $\frac{9n}{m}$ equals

A) 9 B) 2 C) 3 D) 6

55. Let A and B are non-singular matrices of order 3 such that $|A| = 5$ and $A^{-1}B^2 + AB = 0$, then $A^2 |A^2| - \text{adj}\,(\text{adj}\,B)$ is equal to

A) null matrix B) $25A^2 - 5B$ C)$25A^2$ D) $50A^2$

56. Number of points of non-differentiability of $f(x) = ||x|-1| + |\cos \pi x|;\ -2 < x < 2$ is

A) 7 B) 6 C) 5 D) 4

57. If $C_0, C_1, \ldots\ldots, C_{2012}$ are binomial coefficients in the expansion of $(1+x)^{2012}$ and $a_0, a_1, \ldots\ldots, a_{2012}$ are real numbers in arithmetic progression then value of $a_0C_0 - a_1C_1 + a_2C_2 - a_3C_3 + \ldots.. + a_{2012}C_{2012}$ is a

A) Even Number B) Odd number C) Natural number D) Prime number

58. If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are three mutually perpendicular unit vectors and $\vec{d}$ is a unit vector which makes equal angles with $\vec{a}$, $\vec{b}$ and $\vec{c}$ then the value of $|\vec{a}+\vec{b}+\vec{c}+\vec{d}|^2$

A) $4+2\sqrt{2}$ B) $4+2\sqrt{3}$ C) $2+\sqrt{5}$ D) $3+\sqrt{5}$

59. Let the area enclosed by the curve $y = 1 - x^2$ and the line $y = a$, where $0 \le a < 1$, be represented by $A(a)$. If $\frac{A(0)}{A\left(\frac{1}{2}\right)} = k$, then

A) $1 < k < \frac{3}{2}$ B) $\frac{3}{2} < k < 2$ C) $2 < k < \frac{5}{2}$ D) $\frac{5}{2} < k < 3$

60. If system of linear equations $(a-1)x + z = \alpha, x + (b-1)y = \beta$ and $y + (c-1)z = \gamma$ where $a, b,$ $c \in I$ and $\alpha, \beta, \gamma \in R$, does not have a unique solution, then maximum possible value of

APEX JEE-MAIN 2025

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$|a+b+c|$ is

A) 5 B) 1 C) 3 D) 4

61. The modulus of the complex number z such that $|z+3-i|=1$ and $\arg(z)=\pi$ is equal to

A) 1 B) 2 C) 3 D) 4

62. Let A is a symmetric and B is skew symmetric matrix such that $A-B=\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}$ then $|A|$ is

A) $-\frac{3}{4}$ B) $-\frac{1}{4}$ C) $-\frac{11}{4}$ D) $-\frac{9}{4}$

63. A bag contains 16 coins of which two are counter feet with heads on both sides, the rest are fair coins. One coin is selected at random from the bag and tossed, then the probability of getting a head is

A) 9/16 B) 11/16 C) 5/9 D) 3/4

64. If $f(x)=\cos x+\sin x$ and $g(x)=x^2-1$, then $g(f(x))$ is injective in the interval

A) $\left[0, \frac{\pi}{2}\right]$ B) $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$ C) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ D) $[0, \pi]$

65. $\int \frac{dx}{(4\sin x-5\cos x)^2}=$

A) $\frac{1}{5(4\tan x-5)}+C$ B) $\frac{-1}{5(4\tan x-5)}+C$

C) $\frac{1}{4(4\tan x-5)}+C$ D) $\frac{-1}{4(4\tan x-5)}+C$

66. Let $y=y(x)$ satisfies the differential equation $\frac{dy}{dx}=xy^2+x-y^2\sin x-\sin x$, $y(0)=0$, then $y(x)$ is given by

APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

A) $2y=\tan\left(\frac{x^2}{2}+\cos x-1\right)$ B) $y=\tan\left(\frac{x^2}{2}+\cos x-1\right)$

C) $y=\tan\left(\frac{x^2}{2}+\cos x+1\right)$ D) $2y=\tan\left(\frac{x^2}{2}+\cos x+1\right)$

67. Locus of mid points of all chords of hyperbola $3x^2-2y^2+6x-4y=0$ having slope 2 is

A) $3x+4y=-1$ B) $3x-4y=-1$ C) $3x-4y=1$ D) $3x+4y=1$

68. The half of length of the latus rectum of a parabola , whose vertex and focus are on the positive x-axis at a distance R and M(>R) respectively from the origin is

A) $4(M+R)$ B) $2(M+R)$ C) $4(M-R)$ D) $2(M-R)$

69. A variable circle passes through the fixed point $A(p,q)$ and touches x-axis. The locus of the other end of the diameter through A is

A) $(x-p)^2=4qy$ B) $(x-q)^2=4py$ C) $(y-p)^2=4qx$ D) $(y-q)^2=4px$

70. A line makes the same angle θ with each of the x and z axis and the angle β with y axis is such that $\sin^2\beta=3\sin^2\theta$, then $\sec^2\theta$ is equals to

A) $\frac{5}{3}$ B) $\frac{5}{2}$ C) $\frac{3}{2}$ D) 2

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

71. If the function $f(x)=\begin{cases}1+\sin\frac{\pi x}{2} & , \quad x\le 1\\ ax+b & , \quad 1<x<3\\ 6\tan\frac{\pi x}{12} & , \quad 3\le x<6\end{cases}$ is continuous in the interval $(-\infty,\ 6)$

then value of $2a-b$ is

APEX JEE-MAIN 2025

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72. If the area bounded by curve $x + |y| = 1$ and the y-axis is k, then $4k$ is equals to

73. If a, b, c are in A.P. and one root of $ax^2 + bx + c = 0$ is 2 then other root is β then the value of $\frac{-4\beta}{5}$ is

74. The integral $\int_2^4 \frac{\log x^2}{\log x^2 + \log\left(36 - 12x + x^2\right)}\,dx$ is equal to

75. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three vectors such that $|\vec{a}| = \sqrt{3}$, $|\vec{b}| = 5$, $\vec{b} \cdot \vec{c} = 10$ and the angle between $\vec{b}$ and $\vec{c}$ is $\frac{\pi}{3}$. If $\vec{a}$ is perpendicular to the vector $\vec{b} \times \vec{c}$, then $\left|\vec{a} \times \left(\vec{b} \times \vec{c}\right)\right|$ is equal to _____.

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